

Student Number: _____

St Catherine's School



2016 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Extension 2 Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Pages 3 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 7 – 14

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section.

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

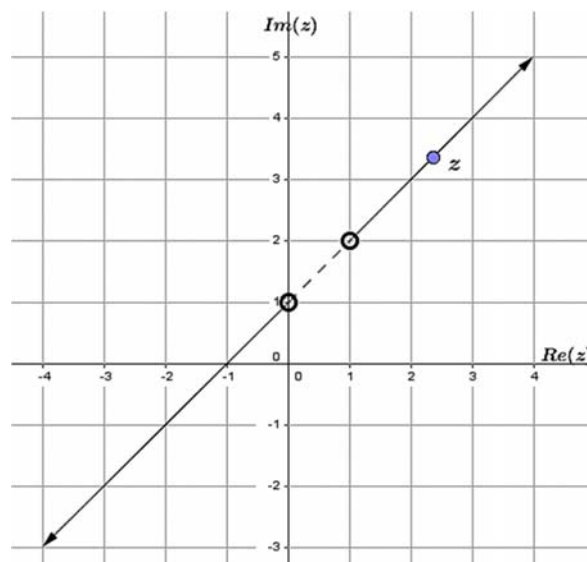
Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which of the following is the correct expansion of $\int \sin^3 x \, dx$?
- (A) $\frac{1}{3} \cos^3 x - \cos x + c$
- (B) $\frac{1}{3} \cos^3 x + \cos x + c$
- (C) $\frac{1}{3} \sin^3 x - \sin x + c$
- (D) $\frac{1}{3} \sin^3 x + \sin x + c$
- 2 If α, β and γ are the roots of $4x^3 - 6x^2 + 11x - 5 = 0$ then the polynomial equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ is
- (A) $12x^3 + 9x^2 - 16x + 2 = 0$
- (B) $3x^3 - 7x^2 + 18x - 11 = 0$
- (C) $5x^3 - 11x^2 + 6x - 4 = 0$
- (D) $2x^3 - 3x^2 - 22x + 10 = 0$
- 3 Six people are divided into three groups of two. The number of different ways this can be done is
- (A) 90
- (B) 45
- (C) 30
- (D) 15

4 The directrices of the hyperbola $\frac{y^2}{9} - \frac{x^2}{16} = 1$ are

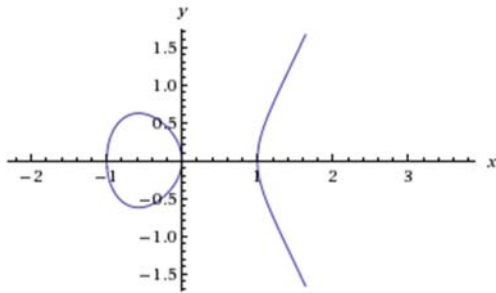
- (A) $x = \pm \frac{9}{5}$
- (B) $y = \pm \frac{9}{5}$
- (C) $y = \pm 5$
- (D) $x = \pm 5$

5 Which of the following defines the locus of the complex number z sketched in the diagram below



- (A) $\arg\left(\frac{z-i}{z-1-2i}\right) = \pi$
- (B) $\arg(z+i) = \arg(z-1-2i)$
- (C) $\arg(z-i) = \arg(z-1-2i)$
- (D) $\arg\left(\frac{z+i}{z-1-2i}\right) = \pi$

- 6 The diagram below shows the graph of $y^2 = f(x)$



Which expression best represents the function $f(x)$?

- (A) $x^2(x - 1)$
 - (B) $x^2(1 - x)$
 - (C) $x(x^2 - 1)$
 - (D) $x(1 - x^2)$
- 7 The complex number z lies on the curve $|z - (1 + i)| = 1$.
What is the maximum value of $|z|$?
- (A) $2 + \sqrt{2}$
 - (B) 2
 - (C) $\sqrt{2} - 1$
 - (D) $\sqrt{2} + 1$
- 8 What is the gradient of the tangent to the curve $-8x^2 + y^2 + 2y = 0$ at the point $(1, 2)$?
- (A) 2
 - (B) $\frac{8}{3}$
 - (C) -1
 - (D) $\frac{4}{5}$

- 9 Without evaluating the integrals, which of the following will give an answer of zero?

(A) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin^3 \theta + 1}{\sin^2 \theta} d\theta$

(B) $\int_{-1}^1 (x^2 - 1)(1 - x^2)^3 dx$

(C) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x \cos x dx$

(D) $\int_{-3}^3 |x^2 - 9| dx$

- 10 Given that $\int \sec^n x dx = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} \int \sec^{n-2} x dx$, then

$$\int_0^{\frac{\pi}{4}} \sec^4 x dx =$$

(A) $\frac{4}{3}$

(B) 1

(C) $\frac{5}{6}$

(D) $\frac{6+4\sqrt{2}}{9}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) If $z = 1 - i\sqrt{3}$ and $w = 1 + i$, 2
i) Express z and w in modulus argument form, 2
ii) find in modulus–argument form the complex number $\frac{z^2}{w^3}$. 2

- (b) If $(1 + i)^n = x + iy$, show that $x^2 + y^2 = 2^n$ 3

- (c) i) The polynomial $P(x) = x^4 - 2x^3 - 3x^2 + ax + b$ has a double root at $x = 2$. Show that $a = 4$ and $b = 4$. 2

- ii) Factorise $P(x)$ fully. 2

- (d) i) Use the substitution $t = \tan \frac{x}{2}$ to evaluate 3

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} dx$$

- ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x} dx$ 1

Question 12 (15 marks)

(a) Find the locus of Z : $|z + 1| < |z|$ 2

(b) i) Consider the expansion of $(\cos \theta + i \sin \theta)^5$.
By writing each of $\sin 5\theta$ and $\cos 5\theta$ in terms of $\cos \theta$ and $\sin \theta$, Show that 2

$$\tan 5\theta = \frac{\tan^5 \theta - 10 \tan^3 \theta + 5 \tan \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

ii) Find the values of θ , for which $\tan \theta$ is a solution of the equation 2
 $x^4 - 10x^2 + 5 = 0$

iii) By solving the equation $x^4 - 10x^2 + 5 = 0$, find the exact values of 3
 $\tan \frac{\pi}{5}$ and $\tan \frac{2\pi}{5}$

(c) i) By writing $\frac{(2x-1)(x+1)}{x-1}$ in the form $mx + b + \frac{a}{x-1}$, find the equation of the oblique 2
asymptote of $y = \frac{(2x-1)(x+1)}{x-1}$.

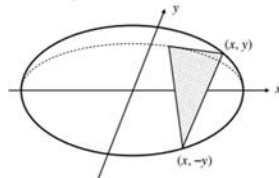
ii) Show that the turning points are (0,1) and (2,9). 2

iii) Hence sketch the graph of $y = \frac{(2x-1)(x+1)}{x-1}$, clearly indicating the intercepts, the 2
asymptotes and the turning points.

Question 13 (15 marks)

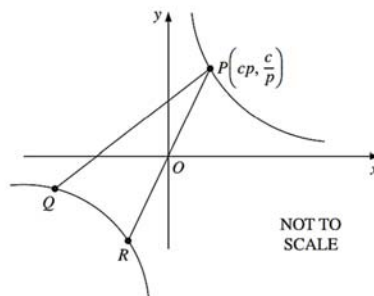
- (a) The base of a solid is in the shape of an ellipse with equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

Sections parallel to the y-axis are equilateral triangles, with one side sitting in the base of the solid, as shown in the diagram.



- i) Show that the volume of the solid is given by $V = \frac{8\sqrt{3}}{9} \int_0^3 (9 - x^2) dx$
- ii) Hence find the volume of the solid.

(b)



$P: (cp, \frac{c}{p})$, where $p \neq \pm 1$ is a point on the hyperbola $xy = c^2$.

The normal to the Hyperbola at P meets it again at the point Q.

The line through P and the origin meets the second branch of the hyperbola at R.

You are given that the equation of the normal at P is

$$py - c = p^3(x - cp) \quad \text{Do not prove this.}$$

- i) Show that if the point Q is $(cq, \frac{c}{q})$, then $q = -\frac{1}{p^3}$
- ii) Show that the coordinates of R is $(-cp, -\frac{c}{p})$.
- iii) Show that angle QRP is a right angle.

Question 13 continues on the next page

(c) Find $\int \frac{\ln x}{x^2} dx$ **2**

(d) i) Find the values of a , b and c such that **2**

$$\frac{3x^2 + 4x + 11}{(x+1)(x^2 + 4)} = \frac{a}{x+1} + \frac{bx+c}{x^2 + 4}$$

ii) Hence, or otherwise, find **2**

$$\int \frac{3x^2 + 4x + 11}{(x+1)(x^2 + 4)} dx$$

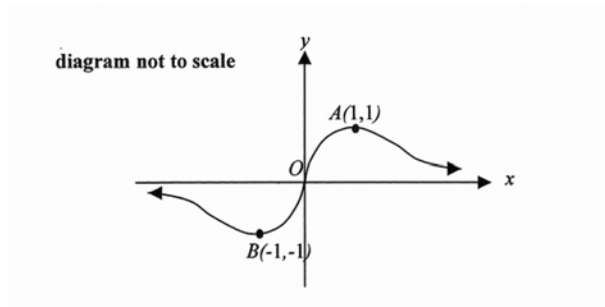
END OF QUESTION 13

Question 14 (15 marks)

- (a) A particle is fired vertically with initial velocity of u metres per second, and is subject both to gravity, g , and air resistance, which is proportional to the square of the speed v .

- i) Show that the equation of motion is given by $\ddot{x} = -g - kv^2$, where k is a constant. 1
- ii) By taking $\ddot{x} = v \frac{dv}{dx}$ and integrating, show that the greatest height H reached by the particle is given by $H = \frac{1}{2k} \ln \frac{g+ku^2}{g}$ 2
- iii) The particle returns to the point of projection. By considering a suitable equation of motion, show that the velocity w , with which it returns to the point of projection is given by $w^2 = \frac{g}{k} (1 - e^{-2kH})$ 3

- (b) In the diagram below the graph of $y = \frac{2x}{1+x^2}$ is sketched showing the turning points A: (1,1) and B: (-1,-1)



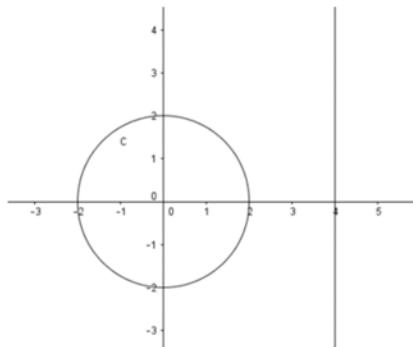
- i) Find the real values of k , for which $\frac{2x}{1+x^2} = kx$ has one solution. 3
- ii) Sketch the graph of $y = \ln \left(\frac{2x}{1+x^2} \right)$ 2

- (c) For the hyperbola $xy = 9$

- i) Show that the coordinates of the foci are $(3\sqrt{2}, 3\sqrt{2})$ and $(-3\sqrt{2}, -3\sqrt{2})$. 2
- ii) Find the equation of the directrices of this hyperbola. 2

Question 15 (15 marks)

- (a) The circle $x^2 + y^2 = 4$ is rotated about the line $x = 4$.



- i) Using the method of cylindrical shells to show that the volume is given by

1

$$V = 4\pi \int_{-2}^2 (4 - x)\sqrt{4 - x^2} \, dx$$

- ii) Hence find the volume of the solid formed.

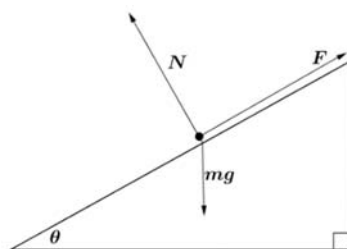
4

- (b) i) A car of mass m Kg is travelling around a circular track of radius R metres, which is inclined at an angle θ to the horizontal.

2

The car has no tendency to side slip. Show that the recommended speed of travel, u metres per second is given by $u^2 = Rg \tan \theta$

- ii)



For a car travelling with a speed of v metres per second, $v \neq u$, show that the sideways frictional force is given by $F = mg \sin \theta - m \frac{v^2}{R} \cos \theta$

2

- iii) If the car is travelling with a speed *one third* the recommended speed, show that the frictional force is given by $F = \frac{8mgu^2}{9\sqrt{u^4 + g^2 R^2}}$

2

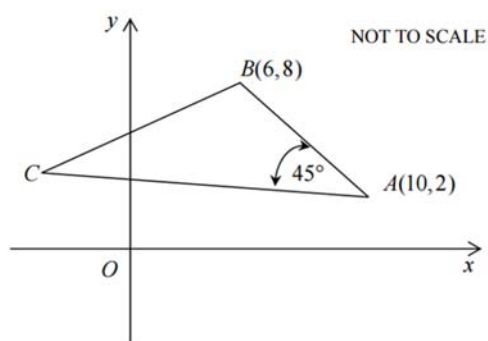
Question 15 continues on the next page

- (c) i) The depth of water in a harbour is 7 metres at low tide and 13 metres at high tide. On a given day, the low tide is at 3AM and high tide is at 9AM. 2
- If the motion of the tide follows Simple Harmonic Motion, show that it can be represented by $x = -3 \cos \frac{\pi}{6} t$, by suitable choice of axes. Explain your choice of axes clearly.
- ii) A ship requires 11.5 metres of water to leave the harbour. Find the earliest time the ship can leave the harbour on that day. 2

END OF QUESTION 15

Question 16 (15 marks)

(a)



In the figure above the length of AC is twice the length of AB.

- i) Explain why $\overrightarrow{AB}$ represents the complex number $-4 + 6i$. 1
- ii) Explain why $\overrightarrow{AC}$ represents the complex number $-10\sqrt{2} + 2\sqrt{2}i$. 2
- iii) Find the complex number C represents. 1

- (b) i) Show that $\int x^2 \sqrt{1-x^3} dx = -\frac{2}{9} \sqrt{(1-x^3)^3} + c$ 1

- ii) Let $I_n = \int_0^1 x^n \sqrt{1-x^3} dx$ for $n \geq 2$. 3

By writing $x^n \sqrt{1-x^3} = x^{n-2} \times x^2 \sqrt{1-x^3}$, or otherwise, show that

$$I_n = \frac{2n-4}{2n+5} I_{n-3} \text{ for } n \geq 5.$$

- iii) Hence find I_8 2

- (c) The polynomial $f(x) = x^3 + cx + d$ has three distinct real roots and hence two turning points at $x = u$ and $x = v$.

- i) Show that u and v are the roots of the equation $x^2 = -\frac{c}{3}$. 1
- ii) Explain why $f(u) \cdot f(v) < 0$ 1
- iii) Hence or otherwise show that $27d^2 + 4c^3 < 0$. 3

END OF PAPER

Student Number: Solution.

2016 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 2

Multiple Choice Answer Sheet

Completely fill the response circle representing the most correct answer

	A	B	C	D
1.	☒	☐	☐	☐
2.	☐	☐	☒	☐
3.	☐	☐	☐	☒
4.	☐	☒	☐	☐
5.	☐	☐	☒	☐
6.	☐	☐	☒	☐
7.	☐	☐	☐	☒
8.	☐	☒	☐	☐
9.	☐	☐	☒	☐
10.	☒	☐	☐	☐

Q.1.
$$\int \sin^3 x \, dx = \int (1 - \cos^2 x) \sin x \, dx$$

$$= \int \sin x - \int \cos^2 x \sin x \, dx$$

$$= -\cos x + \frac{\cos^3 x}{3} + C \quad \boxed{A}$$

Q.2.
$$4\left(\frac{1}{x}\right)^3 - 6\left(\frac{1}{x}\right)^2 + 11\left(\frac{1}{x}\right) - 5 = 0$$

$$4 - 6x + 11x^2 - 5x^3 = 0 \quad \boxed{C}$$

Q.3.
$$\frac{6c_2 \times 4c_2 \times 2c_2}{3!} = 15 \quad \boxed{D}$$

Q.4.
$$b^2 = a^2(e^2 - 1)$$

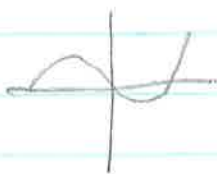
$$e = \frac{5}{3}$$

$$y = \pm 9/e; \pm \frac{3}{5/2}$$

$$= \pm \frac{9}{5} \quad \boxed{B}$$

Q.5.
$$\text{Arg}(z - (1 + 2i)) = \text{Arg}(z - i) \quad \boxed{C}$$

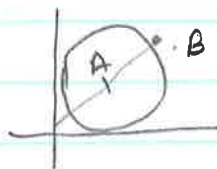
Q.6.



$$y = x(x^2 - 1)$$

$\boxed{C}$

Q.7



$$OA = \sqrt{2} + 1$$

$$AB = 1$$

$$OB \text{ is max; } \sqrt{2} + 1$$

$\boxed{D}$

Q. 8

$$-16x + 2yy' + 2y = 0$$

$$-16 + 4y' + 2y' = 0$$

$$6y' = 16$$

$$y' = \frac{8}{3}$$

[B]

Q. 9.

[C]

$\sin^7 x \cos x$ is an odd fn.

Q. 10

$$\int_0^{\pi/4} \sec^4 x = \left(\frac{1}{3} \tan x \sec^2 x \right)_0^{\pi/4} + \frac{2}{3} \int_0^{\pi/4} \sec^2 x dx$$

$$= \frac{2}{3} + \frac{2}{3} (\tan x)_0^{\pi/4}$$

$$= \frac{4}{3}$$

[A]

Q.11

$$z = 1 - i\sqrt{3}$$



$$(1) z = 2 \operatorname{cis}(-\pi/3)$$

1M

$$w = 1 + i$$

$$= \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

1M

(ii)

$$\left| \frac{z^2}{w^3} \right| = \frac{|z|^2}{|w|^3} = \frac{4}{2\sqrt{2}} = \sqrt{2}$$

$\frac{1}{2}$

$$\operatorname{Arg} \frac{z^2}{w^3} = 2 \operatorname{Arg} z - 3 \operatorname{Arg} w$$

$\frac{1}{2}$

$$= 2\left(-\frac{\pi}{3}\right) - 3\left(\frac{\pi}{4}\right)$$

$$= -1\pi - \frac{3\pi}{4}$$

$$= -\frac{7\pi}{4}$$

1.

$$\frac{z^2}{w^3} = \sqrt{2} \operatorname{cis} \frac{7\pi}{4}$$

$$b) (1+i)^n = x+iy$$

$$1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4} \quad 1M.$$

$$(\sqrt{2} \operatorname{cis} \frac{\pi}{4})^n = x+iy$$

$$x = \sqrt{2}^n \cos n\frac{\pi}{4} \quad y = \sqrt{2}^n \sin n\frac{\pi}{4}$$

1M.

$$x^2 + y^2 = (\sqrt{2}^n)^2 \left(\cos^2 n\frac{\pi}{4} + \sin^2 n\frac{\pi}{4} \right)$$

1M.

$$= 2^n$$

Q. 11c) $P(x) = x^4 - 2x^3 - 3x^2 + ax + b$

$$P'(x) = 4x^3 - 6x^2 - 6x + a$$

$$P(2) = 0 \quad P'(2) = 0 \quad 1M.$$

$$16 - 16 - 12 + 8 + b = 0 \quad b = 4 \quad 1M$$

$$32 - 24 - 12 + a = 0 \quad a = 4 \quad 1M.$$

(11) $(x-2)^2$ is a factor of $P(x)$; find $P(x)$.

$$x^4 - 2x^3 - 3x^2 + 4x + 4 = (x^2 - 4x + 4)(x^2 + 2x + 1)$$

$$= (x-2)^2 (x+1)^2 \quad 1M.$$

d) $\int_0^{\pi} \frac{1 + \cos x}{1 + \sin x} dx$

$$= \int_0^2 \frac{(1+t^2)}{(1+t^2)^2} \cdot \frac{2dt}{1+t^2} \quad (1M)$$

$$= 2 \int_0^2 (1+t^2)^{-2} dt \quad (1M)$$

$$= 2 \left(-\frac{1}{1+t^2} \right)_0^2$$

$$= 1$$

(1)

$$t = \tan \frac{x}{2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$\frac{dx}{dt} = \frac{1}{2} (1+t^2)$$

$$dx = \frac{2dt}{1+t^2}$$

$$\frac{1 + \sin x}{1 + \frac{2t}{1+t^2}}$$

$$= \frac{(1+t)^2}{1+t^2}$$

$$x=0; \quad t=0$$

$$x=\pi \quad t=2$$

(4)

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x} dx &= \int_0^{\frac{\pi}{2}} \frac{\sin x + 1 - 1}{1 + \sin x} dx \\
 &= \int_0^{\frac{\pi}{2}} dx - \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} dx \\
 &= \frac{\pi}{2} - 1 \quad [14]
 \end{aligned}$$

Question 12

c) $|z+1| < |z|$

$x < -\frac{1}{2}$

b) $(\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5$

$$\begin{aligned}
 \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\
 \sin 5\theta &= 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta
 \end{aligned} \quad \left. \vphantom{\begin{aligned} \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\ \sin 5\theta &= 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta \end{aligned}} \right\} \text{IM.}$$

$$\tan 5\theta = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

Divide by $\cos^5 \theta$

$$= \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta} \quad \text{IM.}$$

ii) $x^4 - 10x^2 + 5 = 0$

Let $x = \tan \theta$; if $\tan 5\theta = 0$.

$$5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta = 0$$

$$\tan \theta (\tan^4 \theta - 10 \tan^2 \theta + 5) = 0 \quad [1M]$$

$$\tan 5\theta = 0 \quad ; \quad 5\theta = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\theta = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5} \quad [1M]$$

$\theta = 0$ is a solution to $\tan \theta = 0$

$\therefore \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}$ are the roots to

$$\tan^4 \theta - 10 \tan^2 \theta + 5 = 0 \quad [1M]$$

Equivalently,

$\tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5}$ are the roots

$$\text{of } x^4 - 10x^2 + 5 = 0.$$

iii)

$$x^4 - 10x^2 + 5 = 0$$

$$x^2 = \frac{10 \pm 4\sqrt{5}}{2} = 5 \pm 2\sqrt{5} ; x = \pm \sqrt{5 \pm 2\sqrt{5}} \quad [1M]$$

$$\tan \frac{\pi}{5} > 0 ; \tan \frac{2\pi}{5} > 0 \quad \text{and} \quad \tan \frac{\pi}{5} < \tan \frac{2\pi}{5} \quad [1M]$$

$$\therefore \tan \frac{\pi}{5} = \sqrt{5 - 2\sqrt{5}} \quad [1M]$$

$$\tan \frac{2\pi}{5} = \sqrt{5 + 2\sqrt{5}}$$

(6)

Q.12 c)

$$\frac{(2x-1)(x+1)}{x-1}$$

$$x-1 \overline{) \begin{array}{r} 2x+3 \\ 2x^2+x-1 \\ \underline{2x^2-2x} \\ 3x-1 \\ \underline{3x-3} \\ 2 \end{array}}$$

$$y = 2x+3 + \frac{2}{x-1}$$

1M.

Equation of the oblique asymptote.

$$y = 2x+3.$$

1M.

ii)

$$y' = 2 - 2(x-1)^2$$

1M

Stationary points at $y'=0$ $(x-1)^2=1$

$$x-1 = \pm 1$$

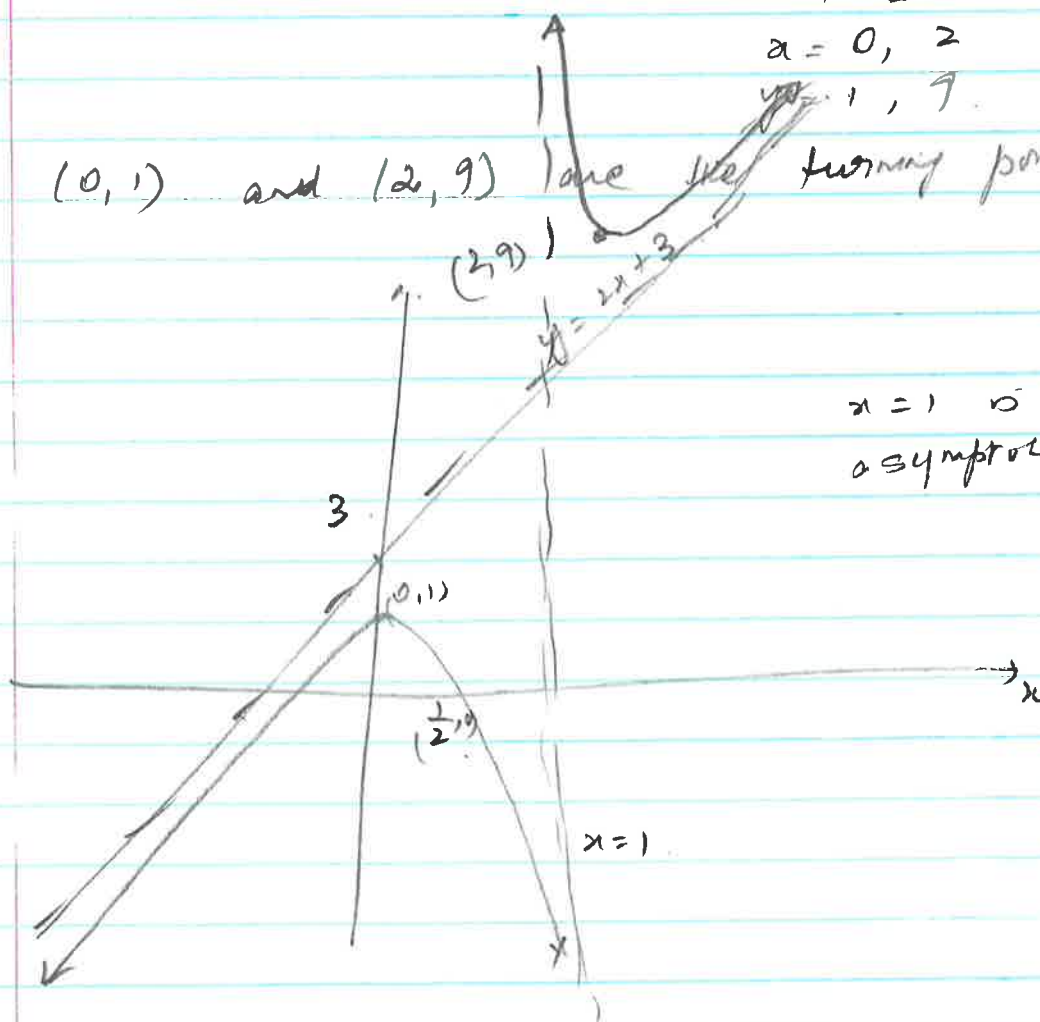
$$x = 0, 2$$

$$y = 1, 9$$

1M

$(0, 1)$ and $(2, 9)$ are the turning points.

iii)

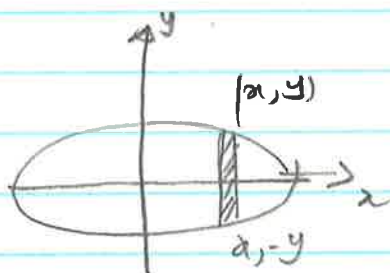


$x=1$ is an asymptote

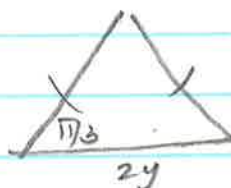
2M

Question 13

a)



A slice



$$\text{Area of a slice} = \frac{1}{2} \cdot 2y \cdot 2y \cdot \sin \frac{\pi}{3}$$

$$= \sqrt{3} y^2$$

$$= \sqrt{3} \left(\frac{4}{9} (9 - x^2) \right)$$

$$\frac{y^2}{4} = 1 - \frac{x^2}{9}$$

$$y^2 = \frac{4}{9} (9 - x^2)$$

$$\Delta v = \sqrt{3} y^2 \Delta x$$

$$\Delta v = \frac{4\sqrt{3}}{9} (9 - x^2) \Delta x$$

(14)

$$v = \frac{4\sqrt{3}}{9} \int_{-3}^3 (9 - x^2) dx = \frac{8\sqrt{3}}{9} \int_0^3 (9 - x^2) dx$$

(14)

$$v = \frac{8\sqrt{3}}{9} \left(9x - \frac{x^3}{3} \right)_0^3$$

$$= \frac{8\sqrt{3}}{9} \times 18$$

$$= 16\sqrt{3}$$

(14)

(b) $xy = c^2$

$$py - c = p^3 (x - cp)$$

passes through $(cp, \frac{c}{p})$

$y' = p$

$\frac{1}{cp}$

$\frac{1}{p^2}$

(8)

$$p \cdot \frac{c}{q} - c = p^3 (cq \cdot cp)$$

$$\cancel{c} \frac{(p-q)}{q} = cp^3 (q-p)$$

1M.

$$\frac{1}{q} = -p^3$$

($p \neq q$)

$$q = -\frac{1}{p^3}$$

1M.

iii)

Equation of OP : $y = \frac{c/p}{cp} x$

$$y = \frac{1}{p^2} x$$

1M.

we have $xy = c^2$

$$x \cdot \frac{x}{p^2} = c^2$$

$$x^2 = p^2 c^2$$

$$x = \pm pc$$

$$x = -cp$$

$$y = -\frac{cp}{p^2} = -\frac{c}{p}$$

$$\therefore R: (-cp, -c/p)$$

1M

iii)

$$m_{QR} = \frac{\frac{c}{q} + \frac{c}{p}}{cq + cp} = \frac{1}{pq}$$

$$m_{PR} = \frac{2c/p}{2cp} = \frac{1}{p^2}$$

$$m_{QR} \times m_{PR} = \frac{1}{pq \times p^2} = \frac{1}{p^3 q} = \frac{-q}{p^3 q} = -1$$

$$(q = -\frac{1}{p^3})$$

$\therefore \angle QRP$ is a right angle

⑦

$$c) \int \frac{\ln x}{x^2} dx$$

$$= -\frac{\ln x}{x} + \int \frac{1}{x} \cdot \frac{1}{x} dx$$

$$= -\frac{\ln x}{x} - \frac{1}{x} + C$$

$$u = \ln x \quad v' = x^{-2}$$

$$u' = \frac{1}{x} \quad v = -\frac{1}{x}$$

$$\int u v' = uv - \int u' v$$

$$d) \frac{3x^2 + 4x + 11}{(x+1)(x^2+4)} = \frac{a}{x+1} + \frac{bx+c}{x^2+4}$$

$$3x^2 + 4x + 11 = a(x^2+4) + (bx+c)(x+1)$$

$$\begin{cases} 3 = a + b \\ 4 = b + c \\ 11 = 4a + c \end{cases} \quad \begin{cases} c - a = 1 \\ c + 4a = 11 \\ 5a = 10 \end{cases} \quad \begin{cases} \underline{b=1} \\ \underline{c=3} \end{cases}$$

$$\underline{a=2}$$

[2M]

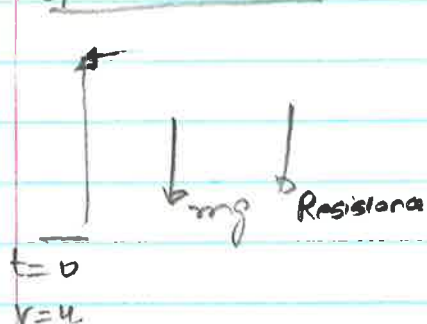
$$\therefore \frac{3x^2 + 4x + 11}{(x+1)(x^2+4)} = \frac{2}{x+1} + \frac{x+3}{x^2+4}$$

$$\int \frac{3x^2 + 4x + 11}{(x+1)(x^2+4)} dx = \int \frac{2}{x+1} dx + \int \frac{x}{x^2+4} dx + \int \frac{3}{x^2+4} dx$$

$$= 2 \ln(x+1) + \frac{1}{2} \ln(x^2+4) + \frac{3}{2} \arctan \frac{x}{2} + C$$

[2M]

Question 14



$$R \propto v^2$$

$$R = mkv^2$$

Equation of motion

$$ma = -mg - mkv^2$$

$$\therefore a = -g - kv^2 \quad (1)$$

(ii)

$$v \frac{dv}{dx} = -g - kv^2$$

$$\int \frac{v dv}{g + kv^2} = - \int dx$$

$$\frac{1}{2k} \ln(g + kv^2) = -x + c$$

$x=0; v=u \quad \therefore c = \frac{1}{2k} \ln(g + ku^2)$ 14

$$\therefore x = \frac{1}{2k} (\ln(g + ku^2) - \ln(g + kv^2))$$

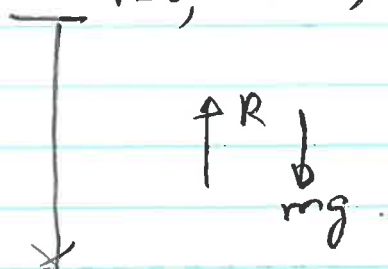
$$= \frac{1}{2k} \ln \frac{g + ku^2}{g + kv^2}$$

$v=0; x=H \quad \therefore H = \frac{1}{2k} \ln \frac{g + ku^2}{g}$ 14

(iii) Downward motion follows a different equation of motion; Reset:

$v=0; a=0; t=0$

$$ma = mg - mkv^2$$



$$a = g - kv^2 \quad (14)$$

$$v \frac{dv}{dx} = g - kv^2$$

(iii)

$$\int \frac{v dv}{g - kv^2} = \int dx$$

$$-\frac{1}{2k} \ln(g - kv^2) = x + C$$

$$x = 0; \quad v = 0 \quad \therefore C = -\frac{1}{2k} \ln g$$

$$\therefore x = \frac{1}{2k} (\ln g - \ln(g - kv^2))$$

$$= \frac{1}{2k} \ln \frac{g}{g - kv^2}$$

[11]

$$x = H; \quad v = w$$

$$H = \frac{1}{2k} \ln \frac{g}{g - kw^2}$$

[12]

$$2kH = \ln \frac{g}{g - kw^2}$$

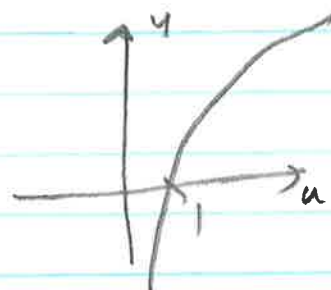
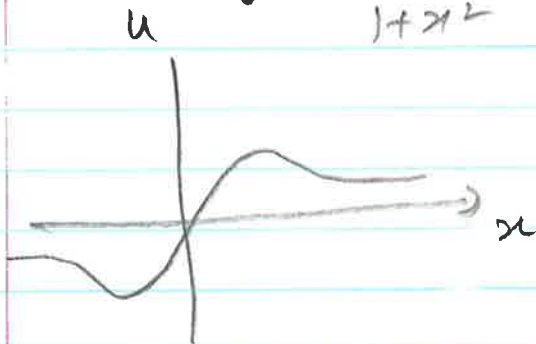
$$e^{-2kH} = \frac{g - kw^2}{g}$$

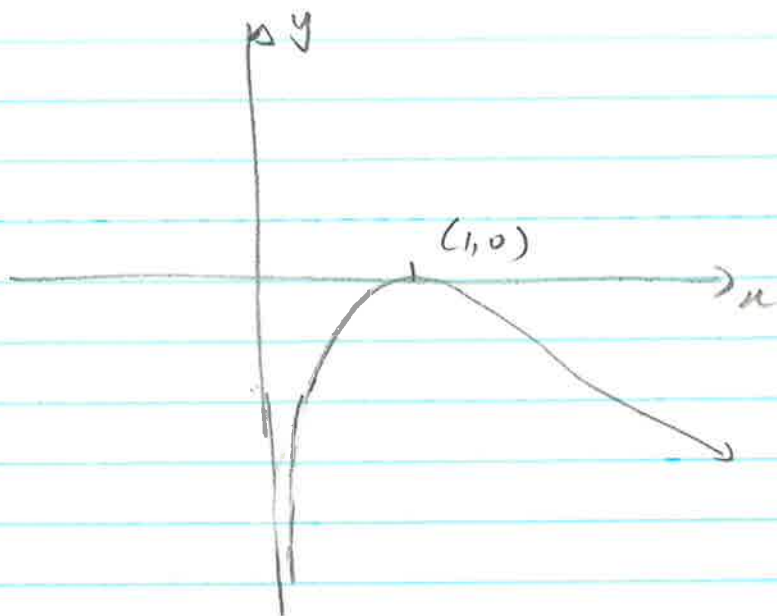
$$\frac{kw^2}{g} = 1 - e^{-2kH}$$

$$w^2 = \frac{g}{k} (1 - e^{-2kH})$$

[13]

b) ⑪ $y = \frac{2x}{1+x^2}$





Notes: (not tested)

$y = \ln \frac{2x}{1+x^2}$ is defined when $\frac{2x}{1+x^2} > 0$; when $x > 0$

$$\frac{2x}{1+x^2} = 1; x = 1; \ln 1 = 0$$

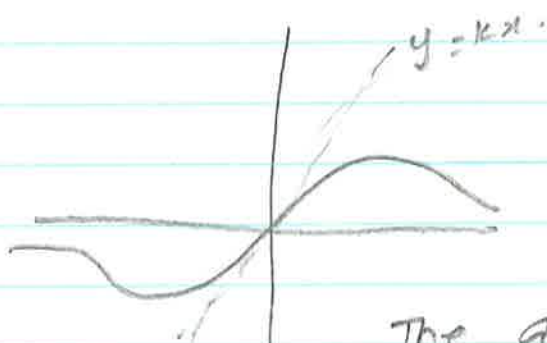
$$0 < x < 1 \quad \frac{2x}{1+x^2} < 1; \ln \frac{2x}{1+x^2} < 0$$

$$x > 1 \quad \frac{2x}{1+x^2} < 1 \quad \ln \frac{2x}{1+x^2} < 0$$

$$x \rightarrow 0 \quad \frac{2x}{1+x^2} \rightarrow 0 \quad \ln \frac{2x}{1+x^2} \rightarrow -\infty$$

$$x \rightarrow \infty \quad \frac{2x}{1+x^2} \rightarrow 0 \quad \ln \frac{2x}{1+x^2} \rightarrow -\infty$$

①



$$y' = \frac{(x^2+1)(2) - 2x(2x)}{(1+x^2)^2}$$

$$y' \big|_{x=0} = 2 \quad \boxed{1M}$$

The gradient of the tangent at (0,0) is 2.

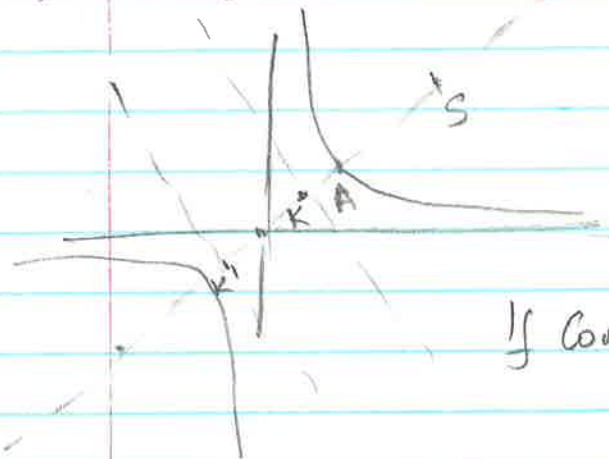
$\frac{2x}{1+x^2} = kx$ has one solution;

$$k \leq 0; k \geq 2 \quad \boxed{1M} \quad \boxed{1M}$$

⑬

c) $xy = 9$

Note $e = \sqrt{2}$



Note $OS = OA \times \sqrt{2}$
 $= \sqrt{3^2 + 3^2} \cdot \sqrt{2}$
 $= 3\sqrt{2} \cdot \sqrt{2}$
 $= 6$ (1M)

If coords. of S: (s, s)
 $s^2 + s^2 = 6^2$
 $s = \pm 3\sqrt{2}$

foci: $(3\sqrt{2}, 3\sqrt{2})$ and $(-3\sqrt{2}, -3\sqrt{2})$ (1M)

② K: where $y = x$ meets the directrix (in quad. I)

$OK = \frac{OA}{e} = \frac{3\sqrt{2}}{\sqrt{2}} = 3$

if $K: (k, k)$ $2k^2 = 9$
 $k = \pm \frac{3}{\sqrt{2}}$

$K: (\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}})$ $K': (-\frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}})$ (1M)

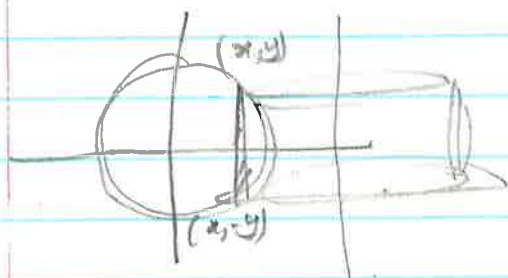
Eqn. of directrix

$y - \frac{3}{\sqrt{2}} = -1(x - \frac{3}{\sqrt{2}})$ (1M)

$x + y = \frac{6}{\sqrt{2}}$ and also $x + y = -\frac{6}{\sqrt{2}}$

Question 15

a)



$$x^2 + y^2 = 4$$

$$y = \pm \sqrt{4 - x^2}$$

$$\Delta V = 2\pi (4 - x) (2y) \Delta x$$

$$= 4\pi (4 - x) \sqrt{4 - x^2} \Delta x$$

$$V = 4\pi \int_{-2}^2 (4 - x) \sqrt{4 - x^2} dx$$

(1M)

(ii) $V = 16\pi \int_{-2}^2 \sqrt{4 - x^2} dx - 4\pi \int_{-2}^2 x \sqrt{4 - x^2} dx$

$$\int_{-2}^2 x \sqrt{4 - x^2} dx = 0$$

$x\sqrt{4 - x^2}$ being an odd fn.

(1M)

$$\therefore V = 16\pi \int_{-2}^2 \sqrt{4 - x^2} dx$$

$$= 16\pi \int_{-2}^2 \sqrt{4 - x^2} dx$$

$$= 16\pi \left(\frac{\pi \cdot (2)^2}{2} \right)$$

$$= 32\pi^2$$

$$x = 2 \sin \theta$$

$$dx = 2 \cos \theta d\theta$$

$$x = 0; \theta = 0$$

$$x = 2; \theta = \frac{\pi}{2}$$

OR

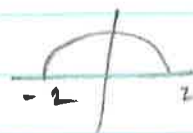
OR

$$V = 32\pi \int_{-2}^2 \sqrt{4 - x^2} dx \quad (\text{even fn.})$$

$$= 32\pi \times 4 \int_0^{\pi/2} \cos^2 \theta d\theta = 64\pi \int_0^{\pi/2} (1 + \cos 2\theta) d\theta$$

$$= 64\pi \left(\theta + \frac{\sin 2\theta}{2} \right) \Big|_0^{\pi/2}$$

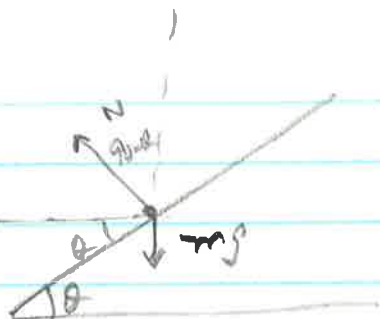
$$= 32\pi^2$$



(15)

15-b)

i)



Resolving forces N and mg horizontally and vertically.

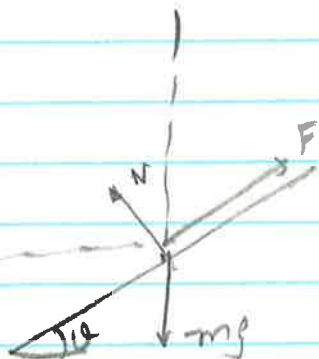
$$F = 0$$

$$N \cos \theta = mg$$

$$N \sin \theta = \frac{mv^2}{R}$$

$$\tan \theta = \frac{v^2}{Rg} \quad v^2 = Rg \tan \theta$$

ii)



Resolve F , N , mg horizontally and vertically,

$$N \cos \theta + F \sin \theta = mg \quad \text{--- (1)}$$

$$N \sin \theta - F \cos \theta = \frac{mv^2}{R} \quad \text{--- (2)}$$

(14)

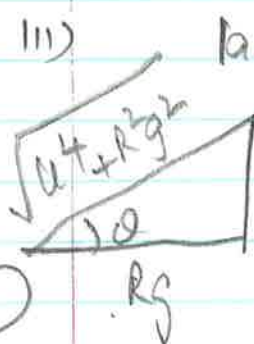
eliminate N : (1) $\times \sin \theta$ - (2) $\times \cos \theta$

$$F \sin^2 \theta + F \cos^2 \theta = mg \sin \theta - \frac{mv^2}{R} \cos \theta$$

$$F = mg \sin \theta - \frac{mv^2}{R} \cos \theta$$

(14)

iii)



$$\tan \theta = \frac{u^2}{Rg}$$

$$\sin \theta = \frac{u^2}{\sqrt{u^4 + R^2 g^2}}$$

$$\cos \theta = \frac{Rg}{\sqrt{u^4 + R^2 g^2}}$$

(14)

$$\therefore F = mg \frac{u^2}{\sqrt{u^4 + R^2 g^2}} - \frac{mv^2}{R} \cdot \frac{Rg}{\sqrt{u^4 + R^2 g^2}}$$

$$v = \frac{u}{3}$$

$$= \frac{mg u^2}{\sqrt{u^4 + R^2 g^2}} - m \frac{u^2}{9} \cdot \frac{g}{\sqrt{u^4 + R^2 g^2}}$$

$$= \frac{8mg u^2}{\sqrt{u^4 + R^2 g^2}}$$

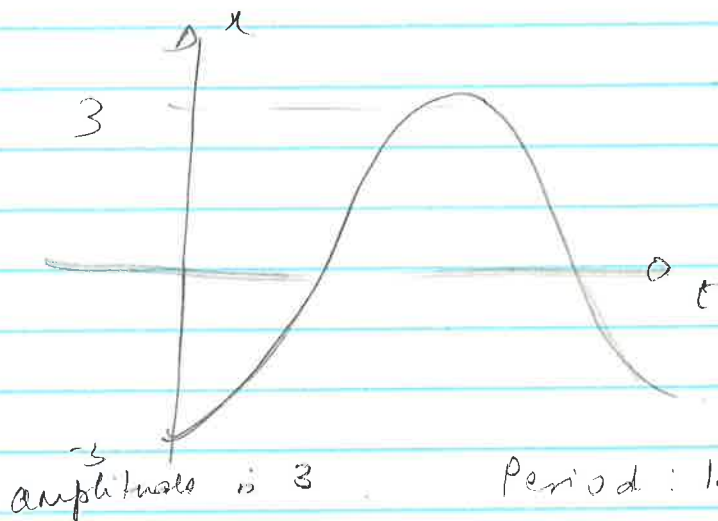
(14)

(16)

c)

Set $x=0$ at 10m.

$t=0$ 3AM ; $x=3$; 13m
 $x=-3$; 7m.



Period : 12 hrs ;

$$\frac{2\pi}{T} = \frac{2\pi}{12}$$

$$\omega = \frac{\pi}{6} \quad (1M)$$

$$\therefore x = -3 \cos \frac{\pi}{6} t \quad (1M)$$

ii) $x = 1.5$ m at 11.5 m after

$$1.5 = -3 \cos \frac{\pi}{6} t$$

$$\cos \frac{\pi}{6} t = -\frac{1}{2}$$

$$\frac{\pi}{6} t = \frac{2\pi}{3}$$

$$t = 4$$

the earliest time is 3AM + 4h = 7AM.

Question 16

a) $\vec{AB} = \vec{OB} - \vec{OA}$
 $= (6+8i) - (10+2i)$
 $= -4 + 6i$

1M.

$$\vec{AC} = \vec{AB} \times 2 \cos \frac{\pi}{4} \quad (1M)$$
$$= (-4+6i)(\sqrt{2} + i\sqrt{2})$$

$$= (-4\sqrt{2} - 6\sqrt{2}) + i(6\sqrt{2} - 4\sqrt{2})$$
$$= -10\sqrt{2} + 2\sqrt{2}i \quad (1M)$$

$$\vec{OC} = \vec{OA} + \vec{AC}$$
$$= (10+2i) - 10\sqrt{2} + 2\sqrt{2}i$$
$$= (10 - 10\sqrt{2}) + i(2 + 2\sqrt{2})$$

b) $\int x^3 \sqrt{1-x^3} dx = -\frac{1}{3} \int (1-x^3)^{\frac{1}{2}} (-3x^2) dx$
 $= -\frac{1}{3} \cdot \frac{2}{3} (1-x^3)^{\frac{3}{2}} + C$
 $= -\frac{2}{9} (1-x^3)^{\frac{3}{2}} + C. \quad (1M)$

$$I_n = \int_0^1 x^n \sqrt{1-x^3} dx$$

$$u = x^{n-2}$$
$$u' = (n-2)x^{n-3}$$

$$v' = x^2 \sqrt{1-x^3}$$
$$v = -\frac{2}{9} (1-x^3)^{\frac{3}{2}}$$

$$\int uv' = uv - \int u'v$$

$$I_n = -\frac{2}{9} \left(x^{n-2} (1-x^3)^{3/2} \right) \Big|_0^1 - \int_0^1 (n-2)x^{n-3} \left(-\frac{2}{9} \right) (1-x^3)^{3/2} dx \quad (1M)$$

$$= \frac{2}{9} (n-2) \int_0^1 x^{n-3} \sqrt{1-x^3} (1-x^3) dx$$

$$= \frac{2}{9} (n-2) \left[\int_0^1 x^{n-3} \sqrt{1-x^3} dx - \int_0^1 x^n \sqrt{1-x^3} dx \right] \quad (1M)$$

$$I_n = \frac{2}{9} (n-2) (I_{n-3} - I_n)$$

$$I_n \left(1 + \frac{2}{9} (n-2) \right) = \frac{2n-4}{9} I_{n-3}$$

$$I_n \cdot \left(\frac{2n+5}{9} \right) = \frac{2n-4}{9} I_{n-3}$$

$$I_n = \frac{2n-4}{2n+5} I_{n-3} \quad (1M)$$

$$I_8 = \frac{12}{21} I_5$$

$$= \frac{12^4}{21^4} \cdot \frac{16^2}{15^2} \cdot I_2 \quad (1M)$$

$$= \frac{8}{35} \left[-\frac{2}{9} \sqrt{1-x^3}^3 \right]_0^1$$

$$= \frac{2 \cdot 8}{135} \times \frac{2}{9} = \frac{16}{315} \quad (1M)$$

c) $f(x) = x^3 + cx + d$

$$f'(x) = 3x^2 + c$$

u & v are roots of $f'(x) = 0$; (since they are the stat. pts)
 $x^2 = -\frac{c}{3}$

ii) 3 distinct real roots

$\Rightarrow$ 3 points of intersection with the x axis
 + 2 stationary points

possible when $(u, f(u))$ and $(v, f(v))$ are on opposite sides of the x axis
 ie $f(u) \cdot f(v) < 0$

iii)

$$f(u) \cdot f(v) = (u^3 + cu + d)(v^3 + cv + d)$$

$$= u^3 v^3 + cu v^3 + dv^3 + cv u^3 + c^2 uv + cvd + du^3 + duv + d^2$$

note $u+v=0$; $uv = -\frac{c}{3}$

$$\therefore f(u) \cdot f(v)$$

$$= \frac{c^3}{27} + cuv(u^2 + v^2) + d(v^3 + u^3) + cd(u+v) + \frac{c^3}{3} + d^2$$

$$= \frac{c^3}{27} + \frac{c^2}{3} \left(-\frac{2c}{3}\right) + 0 + 0 + \frac{c^3}{3} + d^2$$

$$= \frac{4c^3}{27} + d^2 < 0 \quad 4c^3 + 27d^2 < 0$$